

41. There are many people who are single-issue voters, which means that they will vote for (or against!) a candidate based on one issue and one issue only. Suppose a politician wants to earn votes based on single issues: Issue 1, Issue 2, and Issue 3. By publicly supporting issue 1, the politician gains 127 voter but loses 154. By publicly supporting issue 2, the politician gains 350 voters but loses 83. By publicly denouncing Issue 3, the politician gains 306 voters but loses 158. By publicly taking those stances, what is the politician's net gain or loss in number of voters?

3.3 Order of Operations

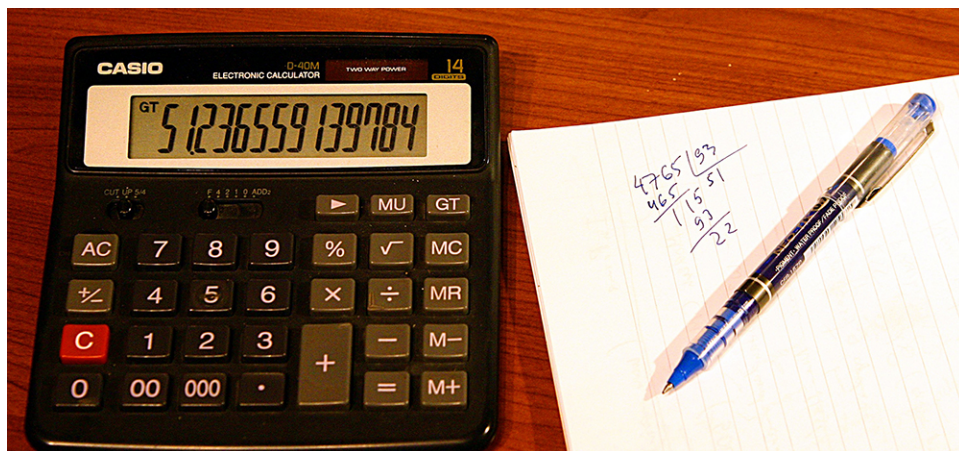


Figure 3.19 Calculators may automatically apply order of operations to calculations. (credit: "Precision" by Leonid Mamchenkov/Flickr, CC BY 2.0)

After completing this module, you should be able to:

1. Simplify expressions using order of operations.
2. Simplify expressions using order of operations involving grouping symbols.

Calculates else sure someone be rules expect explicit we what that needs to need make, we to that them what be calculate to calculated first.

You probably read that sentence and couldn't make heads or tails of it. Seems like it might concern calculations, but maybe it concerns needs? You may even be attempting to unscramble the sentence as you read it, placing words in the order you might expect them to appear in. The reason that the sentence makes no sense is that the words don't follow the order you expect them to follow. Unscrambled, the sentence was intended to be "To be sure that someone else calculates when we expect them to calculate, we need rules that make explicit what needs to be calculated first."

Similarly, when working with math expressions and equations, if we don't follow the rules for **order of operations**, arithmetic expressions make no sense. Just a simple expression would be problematic if we didn't have some rules to tell us what to calculate first. For instance, $4 \times 2^2 + 3 + 5^2$ can be calculated in many ways. You could get 5,184. Or, you could get 80. Or, 96. The issue is that without following a set of rules for calculation, the same expression will give various results. In case you are curious, using the appropriate order of operations, we find $4 \times 2^2 + 3 + 5^2 = 44$.

Simplify Expressions Using Order of Operations

The order in which mathematical operations is performed is a convention that makes it easier for anyone to correctly calculate. They follow the acronym EMDAS:

E	Exponents
M/D	Multiplication and division
A/S	Addition and Subtraction

So, what does EMDAS tell us to do? In an equation, moving left to right, we begin by calculating all the exponents first. Once the exponents have been calculated, we again move left to right, calculating the multiplications and divisions, one at a time. Multiplication and division hold the same position in the ordering, so when you encounter one or the other at this step, do it. Once the multiplications and divisions have been calculated, we again move left to right, calculating the

additions and subtractions, one at a time. Additions and subtractions hold the same position in the ordering, so when you encounter one or the other at this step, do it. (You may have previously learned the order of operations as PEMDAS, with parentheses first; we will add that aspect later on.) We'll explore this as we work an example.

EXAMPLE 3.43**Using Two Order of Operations**

Calculate $21 - 4 \times 13$.

✓ **Solution**

There are no exponents in this expression, so the next operations to check are multiplication and division.

Step 1: Moving left to right, the first multiplication encountered is 4 multiplied by 13. We perform that operation first.

$$\begin{aligned} 21 - 4 \times 13 \\ = 21 - 52 \end{aligned}$$

Step 2: The only operation remaining is the subtraction.

$$21 - 52 = -31.$$

So, $21 - 4 \times 13 = -31$.

 YOUR TURN 3.43

1. Calculate $43 + 18 \times 15$.

EXAMPLE 3.44**Using Two Order of Operations**

Calculate 4×8^3 .

✓ **Solution**

Step 1: Moving left to right, we see there is an exponent. We calculate the exponent first.

$$4 \times 8^3 = 4 \times (8 \times 8 \times 8) = 4 \times 512$$

Step 2: The only operation remaining is the multiplication.

$$4 \times 512 = 2,048$$

So, $4 \times 8^3 = 2,048$.

 YOUR TURN 3.44

1. Calculate $60^2/50$.

EXAMPLE 3.45**Using Three Order of Operations**

Calculate $2 + 3^2 \times 4$.

✓ **Solution**

Step 1: To calculate this, move left to right, and compute all the exponents first. The only exponent we see is the squaring of the 3, so that is calculated first.

$$2 + 3^2 \times 4 = 2 + (3 \times 3) \times 4 = 2 + 9 \times 4$$

Step 2: Since the exponents are all calculated, now calculate all the multiplications and divisions moving left to right. The only multiplication or division present is 9 times 4.

$$2 + 9 \times 4 = 2 + 36$$

Step 3: Moving left to right, perform the additions and subtractions. There is only one such operation, 2 plus 36.

$$2 + 36 = 38$$

$$\text{So, } 2 + 3^2 \times 4 = 38.$$

YOUR TURN 3.45

1. Calculate $3 \times 6^3 - 18$.

Even if the expression being calculated gets more complicated, we perform the operations in the order: EMDAS.

VIDEO

[Order of Operations 1 \(https://openstax.org/r/Order_of_Operations_1\)](https://openstax.org/r/Order_of_Operations_1)

EXAMPLE 3.46

Using Eight Order of Operations

Correctly apply the order of operations to compute the following:

$$4 - 25 \times 6/10 \times 3^2 + 7 \times 2^3.$$

Solution

Step 1: To do so, calculate the exponents first, moving left to right. There are two occurrences of exponents in the expression, 3 squared and 2 cubed.

$$4 - 25 \times 6/10 \times 3^2 + 7 \times 2^3 = 4 - 25 \times 6/10 \times 9 + 7 \times 8$$

Step 2: Now that the exponents are calculated, perform the multiplication and division, moving left to right. The first is the product of 25 and 6.

$$\begin{aligned} 4 - 25 \times 6/10 \times 9 + 7 \times 8 \\ = 4 - 150/10 \times 9 + 7 \times 8 \end{aligned}$$

Step 3: Next is the 150 divided by 10.

$$\begin{aligned} 4 - 150/10 \times 9 + 7 \times 8 \\ = 4 - 15 \times 9 + 7 \times 8 \end{aligned}$$

Step 4: Next is 15 multiplied by 9.

$$\begin{aligned} 4 - 15 \times 9 + 7 \times 8 \\ = 4 - 135 + 7 \times 8 \end{aligned}$$

Step 5: Finally, multiply the 7 and 8.

$$\begin{aligned} 4 - 135 + 7 \times 8 \\ = 4 - 135 + 56 \end{aligned}$$

As all the multiplications and divisions have been calculated, the additions and subtractions are performed, moving left to right.

$$\begin{aligned} 4 - 135 + 56 \\ = -131 + 56 \\ = -131 + 56 \\ = -75 \end{aligned}$$

The computed value is -75 .

YOUR TURN 3.46

1. Correctly apply the order of operations to compute the following:

$$13 + 3^4 \times 8/6 \times 5^2 - 3 \times 4.$$

VIDEO

Order of Operations 2 (https://openstax.org/r/Order_of_Operations_2)

EXAMPLE 3.47

Using Six Order of Operations

Correctly apply the rules for the order of operations to accurately compute the following:

$$10 - 3 \times 5^3/15 + 56/4.$$

Solution

Step 1: Calculate exponents first, moving left to right:

$$\begin{aligned} 10 - 3 \times 5^3/15 + 56/4 \\ = 10 - 3 \times 125/15 + 56/4 \end{aligned}$$

Step 2: Multiply and divide, moving left to right:

$$\begin{aligned} 10 - 3 \times 125/15 + 56/4 \\ = 10 - 375/15 + 56/4 \\ = 10 - 25 + 56/4 \\ = 10 - 25 + 14 \end{aligned}$$

Step 3: Add and subtract, moving left to right:

$$\begin{aligned} 10 - 25 + 14 \\ = -15 + 14 \\ = -1 \end{aligned}$$

YOUR TURN 3.47

1. Correctly apply the rules for the order of operations to accurately compute the following:

$$12/(-4) + 8 \times 9/12 \times 2^3 - 24 \times 25/10.$$

EXAMPLE 3.48

Using Order of Operations

Correctly apply the rules for the order of operations to accurately compute the following:

$$(-8)/2 \times 3 - 9 \times 2^4/12 + 9 \times (-4)^2/2^3.$$

Solution

Step 1: Calculate the exponents first, moving left to right:

$$\begin{aligned}
 &(-8)/2 \times 3 - 9 \times 2^4/12 + 9 \times (-4)^2/2^3 \\
 &= (-8)/2 \times 3 - 9 \times 16/12 + 9 \times 12^2/2^3 \\
 &= (-8)/2 \times 3 - 9 \times 16/12 + 9 \times 144/2^3 \\
 &= (-8)/2 \times 3 - 9 \times 16/12 + 9 \times 144/8
 \end{aligned}$$

Step 2: Multiply and divide, moving left to right:

$$\begin{aligned}
 &= (-8)/2 \times 3 - 9 \times 16/12 + 9 \times 144/8 \\
 &= (-4) \times 3 - 9 \times 16/12 + 9 \times 144/8 \\
 &= (-12) - 9 \times 16/12 + 9 \times 144/8 \\
 &= (-12) - 144/12 + 9 \times 144/8 \\
 &= (-12) - 12 + 9 \times 144/8 \\
 &= (-12) - 12 + 1,296/8 \\
 &= (-12) - 12 + 162
 \end{aligned}$$

Step 3: Add and subtract, moving left to right:

$$= (-12) - 12 + 162 = (-24) + 162 = 138$$

YOUR TURN 3.48

1. Correctly apply the rules for the order of operations to accurately compute the following:

$$3 \times 4^3 \times 7 + 24/6 \times 7^2 - 9/3 \times 8.$$

Using the Order of Operations Involving Grouping Symbols

We have examined how to use the order of operations, denoted by EMDAS, to correctly calculate expressions. However, there may be expressions where a multiplication should happen before an exponent, or a subtraction before a division. To indicate an operation should be performed out of order, the operation is placed inside parentheses. When parentheses are present, the operations inside the parentheses are performed first. Adding the parentheses to our list, we now have PEMDAS, as shown below.

P	Parentheses	
E	Exponents	
M/D	Multiplication and division	(division is just the multiplication by the reciprocal)
A/S	Addition and subtraction	(subtraction is just the addition of the negative)

As said previously, parentheses indicate that some operation or operations will be performed outside the standard order of operation rules. For instance, perhaps you want to multiply 4 and 7 before squaring. To indicate that the multiplication happens before the exponent, the multiplication is placed inside parentheses: $(4 \times 7)^2$.

This means operations inside the parentheses take precedence, or happen before other operations. Now, the first step in calculating arithmetic expressions using the order of operations is to perform operations inside parentheses first. Inside the parentheses, you follow the order of operation rules EMDAS.

EXAMPLE 3.49

Prioritizing Parentheses in the Order of Operations

Correctly apply the rules for the order of operations to accurately compute the following:

$$(10 - 3) \times 5^3.$$

 Solution

Step 1: Perform all calculations within the parentheses before all other operations.

$$(10 - 3) \times 5^3 = 7 \times 5^3$$

Step 2: Since all parentheses have been cleared, move left to right, and compute all the exponents next.

$$7 \times 5^3 = 7 \times 125$$

Step 3: Perform all multiplications and divisions moving left to right.

$$7 \times 125 = 875$$

 YOUR TURN 3.49

1. Correctly apply the rules for the order of operations to accurately compute the following:

$$8 - (25 - 2^2)/7.$$

Be aware that there can be more than one set of parentheses, and parentheses within parentheses. When one set of parentheses is inside another set, do the innermost set first, and then work outward.

 VIDEO

[Order of Operations 3 \(https://openstax.org/r/Order_of_Operations_3\)](https://openstax.org/r/Order_of_Operations_3)

EXAMPLE 3.50

Working Innermost Parentheses in the Order of Operations

Correctly apply the rules for order of operations to accurately compute the following:

$$4 + 2 \times (3^2 - (2 + 5)^2 \times 4)/(3 + 8).$$

 Solution

Step 1: Perform all calculations within the parentheses before other operations. Evaluate the innermost parentheses first. We can work separate parentheses expressions at the same time. The innermost set of parentheses has the $2 + 5$ inside. The $3 + 8$ is in a separate set of parentheses, so that addition can occur at the same time as the $2 + 5$.

$$\begin{aligned} 4 + 2 \times (3^2 - (2 + 5)^2 \times 4)/(3 + 8) \\ = 4 + 2 \times (3^2 - (7)^2 \times 4)/(11) \end{aligned}$$

Step 2: Now that those parentheses have been handled, move on to the next set of parentheses. Applying the order of operation rules inside that set of parentheses, the exponent is evaluated first, then the multiplication, and then the addition.

$$\begin{aligned} 4 + 2 \times (3^2 - 7^2 \times 4)/11 \\ = 4 + 2 \times (3^2 - 49 \times 4)/11 \\ = 4 + 2 \times (-187)/11 \end{aligned}$$

Step 3: Since all parentheses have been cleared, apply the EMDAS rules to finish the calculation.

$$4 + 2 \times (-187)/11 = 4 - 374/11 = 4 - 34 = -30$$

 YOUR TURN 3.50

1. Correctly apply the rules for the order of operations to accurately compute the following:

$$(8 - 6)^2 \times 100 - ((48/6 - 3)^2 - 4 \times 7).$$

 VIDEO

Order of Operations 4 (https://openstax.org/r/Order_of_Operations_4)

Check Your Understanding

12. Which operation has highest precedence?
13. Which is performed first, exponents or addition?
14. Calculate $2 \times 3^2 - 5 \times 8$.
15. What is used to indicate operations that should be performed out of order?
16. Calculate $(4 - 3)^2 + 27 \times 8^2 \div 6^2$.



SECTION 3.3 EXERCISES

1. Which operations have the lowest precedence in order of operations?
2. If many operations have the same precedence in an expression, in what order should the operations be performed?
3. Which operations have the same precedence in order of operations?
4. After all operations in parentheses have been performed, which operations should be performed next?

For the following exercises, perform the indicated calculation.

5. $4 - 5 \times 6$
6. $34 - 10 \times 6$
7. $18 + 8 \times 13$
8. $40 + 12 \times 17$
9. $50 \div 2 + 8$
10. $72 \div 6 + 18$
11. $13 + 4 \times 3^2$
12. $45 - 6 \times 7^3$
13. $6^2 \times 5 - 13 \times 9^2$
14. $14^2 \times 8 - 5 \times 2^4$
15. $450 \div 3^2 - 56 \div 2^2$
16. $1,000 \div 5^3 - 7 \times 8^4$
17. $38 \times 6 - 4 + 5 \times 18 \div 10$
18. $15 \times 7 + 23 - 6 \times 40 \div 24$
19. $600 \div 12 \times 5 + 40 - 6^2$
20. $2 \times 12^3 \div 8 \times 3 - 4^5$
21. $10 \times 6^3 \div 2 \times 5 + 3^2 - 240 \div 8 \times 9$
22. $45 \div 15 \times 6 + 7^2 - 4 \times 15 \times 18 \div 12 \times 3$
23. $(4 + 3) \times 2$
24. $6 \times (12 + 8)$
25. $(14 - 25) \times 2$
26. $(86 - 61) \div 5$
27. $(3 + 2)^3$
28. $(17 - 12)^4$
29. $(45 - 60)^2$
30. $(90 - 101)^3$
31. $(3 \times 4)^3 \div 8 + 5$
32. $(6 - 9)^2 \times 5 - 4$
33. $4 - (6 + 3) \times 5 + 11$
34. $12 + (13 - 6) \times 8 - 130$